

Safe Band: IoT-Based Smart Wearable for Emergency Health Alerts and Monitoring in Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Emergency medical response in Sri Lanka faces significant challenges, including delayed ambulance dispatch, inefficient hospital routing, and a lack of real-time coordination between patients, hospitals, and emergency services. Existing systems largely depend on manual processes, which increase response times and contribute to preventable fatalities during critical situations.

The proposed solution, Safe Band, is a smart wearable system designed to improve emergency healthcare response. This component, the ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification, is activated when a critical health event is detected by the Emergency Detection System (Component 3).

Once triggered, the system captures the patient's real-time GPS location using an Android smartwatch. A machine learning algorithm, specifically K-Nearest Neighbours (KNN) enhanced with Haversine distance computation, is applied to identify the nearest and most suitable hospital based on geographical proximity and operational availability. The patient's critical health data, along with location details, are then transmitted in real time to the selected hospital through the Firebase Realtime Database.

On the hospital side, a web-based dashboard displays the incoming emergency information, enabling healthcare staff to assess the situation instantly. This allows the hospital to dispatch an ambulance with full situational awareness, ensuring faster and more effective emergency response.

The system integrates multiple technologies, including Android smartwatches for biometric and location sensing, a companion mobile application for data transmission, Firebase as a cloud backend, and a hospital-facing web interface.

Expected outcomes of this system include reduced ambulance dispatch times, improved hospital preparedness, and enhanced coordination among emergency stakeholders. Ultimately, it aims to decrease emergency response latency across Sri Lanka. This solution aligns with the United Nations Sustainable Development Goal 3 (Good Health and Well-Being) and supports the country's ongoing Digital Health initiatives.

Keywords: Smart Ambulance Dispatch, Nearest Hospital Identification, K-Nearest Neighbours, IoT Healthcare, Firebase, Emergency Response, GPS Routing, Sri Lanka.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AES	Advanced Encryption Standard
BLE	Bluetooth Low Energy
CSV	Comma-Separated Values
DB	Database
DSR	Design Science Research
ETA	Estimated Time of Arrival
FCM	Firebase Cloud Messaging
FR	Functional Requirement
GPS	Global Positioning System
HIS	Hospital Information System
HR	Heart Rate
HTTP	Hypertext Transfer Protocol
ICT	Information and Communications Technology
IDE	Integrated Development Environment
IoT	Internet of Things
IP	Intellectual Property
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbours
LKR	Sri Lankan Rupee
ML	Machine Learning
MoH	Ministry of Health
MQTT	Message Queuing Telemetry Transport
NFR	Non-Functional Requirement

OCR	Optical Character Recognition
PPG	Photoplethysmography
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDK	Software Development Kit
SDG	Sustainable Development Goal
SLA	Service Level Agreement
SLIIT	Sri Lanka Institute of Information Technology
SO	Specific Objective
SpO2	Peripheral Oxygen Saturation
TLS	Transport Layer Security
UI	User Interface
UN	United Nations
WBS	Work Breakdown Structure
WHO	World Health Organization
Wi-Fi	Wireless Fidelity

Table 1: Abbreviation

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1. INTRODUCTION

1.1 Background and Context

Emergency medical care is one of the most time-critical domains in healthcare. The World Health Organization (WHO) recognizes that each minute of delay in treating a cardiac arrest reduces the probability of survival by 7–10% [1]. In Sri Lanka, emergency healthcare infrastructure remains heavily reliant on manual processes. Citizens dial 1990 (Suwa Seriya), an emergency ambulance service, and dispatch is handled by telephone operators without automated location detection or intelligent hospital routing. This results in significant delays, particularly in urban areas with high traffic density and rural areas with limited hospital infrastructure.

The burden of non-communicable diseases (NCDs) in Sri Lanka is considerable. According to the Ministry of Health, NCDs account for approximately 75% of all deaths in the country, with cardiovascular disease being the leading cause [2]. Patients experiencing acute cardiovascular events, strokes, or sudden drops in blood oxygen saturation require immediate ambulance response and pre-admission hospital preparation, both of which are currently absent in Sri Lanka's emergency ecosystem.

The Safe Band project proposes an IoT-based smart wearable that continuously monitors physiological parameters including heart rate, SpO₂, blood pressure, and body temperature using sensors embedded in an Android-integrated smartwatch. When the Emergency Detection component identifies a critical threshold breach, the system activates a chain of automated responses. This proposal specifically addresses the ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification component, which handles everything from identifying the patient's real-time GPS location to selecting the optimal hospital and initiating an ambulance dispatch.

Globally, systems such as the Emergency Medical Services (EMS) in the United States and the NHS Ambulance Service in the United Kingdom use Computer-Aided Dispatch (CAD) systems that partially automate the process. However, these systems are expensive, require dedicated infrastructure, and are not adapted to low-and-middle-income country (LMIC) contexts such as Sri Lanka [3]. Existing commercial smartwatches such as the Apple Watch and Samsung Galaxy Watch include emergency SOS features but do not integrate with ambulance dispatch systems or hospital management platforms [4].

This system aligns with the United Nations Sustainable Development Goal 3 (SDG 3): Ensure healthy lives and promote well-being for all ages, specifically Target 3.6 (reduce road traffic injuries) and Target 3.d (strengthen health security and emergency preparedness). Additionally, it aligns with SLIIT's research cluster in Smart Systems and IoT.

1.2 Literature Review

A substantial body of research exists on emergency medical dispatch optimisation, IoT-based health monitoring, and machine learning-assisted hospital routing. This section reviews twelve key scholarly sources that directly inform the design of the ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification. The review covers existing research approaches, existing products and services in the domain, and the limitations of current systems that the proposed component aims to address.

Existing Research Approaches

Rajapakse et al. (2022) conducted a national survey of emergency response outcomes in Sri Lanka and found that average ambulance response times in Colombo exceed 18 minutes, compared to the WHO-recommended benchmark of 8 minutes for critical emergencies [5]. Their research highlighted the absence of automated location services and hospital pre-notification as the two most significant systemic bottlenecks in Sri Lanka's emergency medical chain. This finding directly motivates the real-time GPS acquisition and hospital dashboard notification features designed into Component 4 of the Safe Band system.

Agarwal and Mehta (2021) proposed a deep learning model for ambulance routing that integrated traffic density data and hospital capacity metrics, achieving a 22% improvement in route optimisation over baseline Dijkstra pathfinding [6]. Their research demonstrated that incorporating real-time hospital occupancy rates into the routing decision significantly reduces dispatch-to-arrival time and establishes a measurable benchmark for ML-assisted routing improvement. This study validates the core motivation behind Component 4's machine learning approach to hospital identification and provides a target improvement figure against which the system's performance can be evaluated.

Al-Turjman et al. (2020) presented a comprehensive IoT framework for smart ambulance systems using the MQTT protocol for real-time data transmission between wearable sensors and dispatch centres [7]. Their architecture demonstrated the feasibility of sub-second sensor-to-cloud latency using a lightweight publish-subscribe messaging model. This study serves as a foundational design reference for the Safe Band real-time alert pipeline, as the publish-subscribe pattern they propose is architecturally equivalent to the Firebase Realtime Database listener model adopted in Component 4 for receiving emergency triggers and dispatching hospital notifications.

Ketu and Mishra (2021) evaluated K-Nearest Neighbours (KNN) and Random Forest algorithms for hospital selection, using distance, medical specialisation, and real-time bed availability as selection criteria [8]. Their study found that Random Forest outperformed KNN by 15% in precision for multi-criteria hospital selection tasks. While this supports the potential of ensemble ML methods, for proximity-only scenarios which is the primary use case in the initial Safe Band deployment KNN with Haversine distance computation provides sufficient accuracy with significantly lower computational overhead, making it the most appropriate choice for execution within a Firebase Cloud Function environment.

Mehmood et al. (2019) introduced a smart city ambulance dispatch model integrating GIS data with machine learning to identify optimal hospital-ambulance pairings [9]. Their system processed events from detection to dispatch trigger in an average of 3.2 seconds, demonstrating that ML-assisted dispatch can operate well within real-time emergency response constraints. This establishes a clear benchmark for the Safe Band Component 4 end-to-end latency target of under 30 seconds, which encompasses GPS acquisition, KNN computation, Firebase write, and FCM notification delivery a significantly more complex pipeline than the one Mehmood et al. evaluated.

Faulkner et al. (2020) reviewed the accuracy of wearable sensor data in detecting cardiac events, finding that consumer-grade smartwatch ECG sensors achieve up to 87% sensitivity in atrial fibrillation detection [10]. This study supports the reliability of using Android smartwatch sensor inputs as the data source for emergency detection within the Safe Band system. The findings validate the hardware choice made for the overall project and justify Component 4's design assumption that

incoming alert triggers from Component 3 carry sufficient clinical reliability to warrant automated ambulance dispatch initiation.

Li et al. (2020) proposed a reinforcement learning-based dynamic ambulance redeployment model that improved emergency coverage in urban areas by 28% by repositioning idle ambulances to maximise geographic coverage probability [15]. While this approach offers superior long-term optimisation compared to static nearest-hospital identification, the authors acknowledged that effective model training requires several months of historical dispatch data. The Safe Band system, operating in an early deployment phase without such data, adopts a simpler but practically effective proximity-based KNN model as its baseline, with RL-based redeployment identified as a future enhancement.

Existing Products and Services

Lakshmi and Renuka (2022) proposed and validated a Firebase-based real-time notification architecture for hospital emergency rooms, demonstrating that Firebase Cloud Messaging achieves 99.1% delivery reliability with an average notification latency of 420 milliseconds [11]. This existing product architecture is directly adopted in the Safe Band hospital dashboard notification sub-component. The sub-500ms delivery performance confirmed by this study indicates that FCM is not the latency bottleneck in the system GPS acquisition and KNN computation accounts for the dominant share of the 30-second end-to-end target.

Garg et al. (2021) evaluated existing geospatial service platforms specifically Google Maps Platform and OpenStreetMap for accuracy in determining the nearest service provider in South Asian urban environments [12]. Google Maps API was found to be 18% more accurate in distance estimation than OpenStreetMap in dense urban grids, while OpenStreetMap offered better road network coverage in rural zones. This study directly informs the hybrid mapping strategy adopted in Component 4, which uses the Google Maps SDK for the hospital dashboard routing visualisation while maintaining OpenStreetMap as a fallback for rural hospital locations across Sri Lanka's mixed urban-rural geography.

Silva et al. (2023) studied hospital response preparedness in South Asian emergency departments and found that existing pre-arrival patient data sharing systems reduced in-hospital preparation time by an average of 34% [13]. The most impactful data elements identified were patient condition classification, vital signs such as heart rate and oxygen saturation, and GPS-based estimated time of arrival all of which are included in the Component 4 Firebase dispatch notification payload. This study provides direct clinical evidence validating the hospital dashboard notification feature as the highest-impact output of the proposed component.

Nanayakkara et al. (2022) reviewed the adoption of digital health technologies in Sri Lanka and found that 74% of surveyed urban Sri Lankans expressed willingness to use emergency wearable devices [16]. The study identified perceived safety benefit, ease of use, and affordability as the primary adoption drivers factors directly addressed by the Safe Band system's use of existing consumer Android smartwatches rather than proprietary medical-grade hardware. This finding confirms the market feasibility of the overall Safe Band system and provides population-level evidence supporting the commercialisation plan target market assumptions for Component 4.

Limitations of Existing Systems

Despite the advances demonstrated across the reviewed literature, several critical limitations constrain the applicability of existing systems to the Sri Lankan emergency healthcare context. Agarwal and Mehta's (2021) deep learning routing model required dedicated city-level infrastructure and real-time cloud-based traffic APIs that are not natively available in Sri Lanka, making it unsuitable for direct adoption [6]. Similarly, Al-Turjman et al.'s (2020) MQTT-based IoT framework depends on custom broker configuration and dedicated server infrastructure, whereas the Safe Band system requires a managed, low-cost cloud solution accessible without specialist DevOps expertise [7].

Wickramasinghe et al. (2021) identified that only 12% of hospitals in Sri Lanka have a digital bed management system, which directly invalidates the multi-criteria hospital selection model proposed by Ketu and Mishra (2021), since it relies on real-time hospital capacity data feeds that 88% of Sri Lankan hospitals cannot provide [8][14]. This gap necessitates a dispatch model that operates effectively without internal hospital system integration, which is a foundational design constraint of Component 4. The proposed system therefore uses proximity-based KNN as its primary selection criterion, with capacity data integration planned as a future enhancement once hospital digital infrastructure matures.

Li et al.'s (2020) reinforcement learning model, while achieving a 28% improvement in coverage, requires several months of historical dispatch data for training a dataset that does not yet exist in the Sri Lankan context [15]. Mehmood et al.'s (2019) GIS-integrated dispatch model similarly assumes the presence of comprehensive urban GIS infrastructure unavailable outside major Sri Lankan cities [9]. Faulkner et al. (2020) noted that smartwatch ECG sensitivity for non-atrial-fibrillation cardiac events remains below 80%, introducing a false dispatch risk that Component 4 must mitigate through integration with Component 3's confidence scoring and a target false dispatch rate of under 2% [10].

From a geographic platform perspective, Garg et al. (2021) highlighted that Google Maps API accuracy degrades in rural areas while its cost scales with query volume a constraint that requires the OSM fallback strategy and query caching mechanism included in the Component 4 design [12]. Silva et al. (2023) assumed hospitals possess existing electronic health record systems, which Wickramasinghe et al. (2021) confirm are absent in most Sri Lankan government hospitals, requiring Component 4's dashboard to function as a fully standalone notification interface [13][14]. Finally, Nanayakkara et al.'s (2022) market survey was limited to urban areas, leaving rural adoption barriers including mobile network coverage and smartphone literacy as open research questions for future Safe Band deployment studies [16].

Collectively, the reviewed literature confirms the clinical necessity and technical feasibility of an automated, ML-driven ambulance dispatch and hospital notification system for Sri Lanka. However, no existing system integrates wearable-triggered emergency detection, Firebase-mediated real-time communication, KNN-based geospatial hospital identification, and a hospital-facing dashboard notification pipeline within a single, infrastructure-light architecture suited to Sri Lanka's healthcare landscape.

1.3 Research Gap

Despite the body of literature reviewed above, several critical gaps remain unaddressed, particularly within the Sri Lankan healthcare and emergency services context:

- No integrated real-time GPS-to-dispatch pipeline exists in Sri Lanka: Existing systems such as Suwa Seriya (1990) rely on verbal location reporting. No automated mechanism links a patient's real-time device location to ambulance dispatch.
- Context-adapted hospital selection is absent: Global systems assume the availability of real-time hospital capacity APIs, which do not exist in Sri Lanka. A model that identifies the nearest suitable hospital using publicly available geospatial data, hospital type, and offline proximity calculation is required.
- Lack of pre-arrival clinical data sharing: No current system in Sri Lanka transmits patient health parameters to the receiving hospital prior to ambulance arrival, causing delays in in-hospital preparation.
- IoT-to-dispatch integration for wearables has not been implemented locally: While global research demonstrates the technical feasibility of wearable-triggered ambulance dispatch, no such system has been designed, deployed, or evaluated in a Sri Lankan healthcare setting.
- ML models for ambulance dispatch in LMICs are underexplored: Most ML dispatch models are designed for high-income country infrastructure with abundant training data, dedicated sensors, and connected hospital systems. A model designed to operate within resource-constrained environments is absent from literature.

The proposed ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification directly addresses these gaps by delivering a fully automated, GPS-triggered, MLdriven dispatch pipeline that is specifically designed for Sri Lanka's healthcare infrastructure, requiring no real-time hospital API integration and functioning reliably with commercially available Android smartwatch sensors.

2. OBJECTIVE

2.1 Main Objective

To design and implement an ML-Based Smart Ambulance Dispatch System that automatically identifies a patient's real-time GPS location upon critical health event detection, determines the nearest suitable hospital using a machine learning algorithm, transmits patient health data to the hospital dashboard, and triggers ambulance dispatch achieving an end-to-end response initiation time of under 10 seconds from alert generation.

2.2 Specific Objectives

- To develop a real-time GPS location acquisition module that retrieves and validates the patient's current geographical coordinates from the Android smartwatch upon receipt of a critical alert from the Emergency Detection System component.
- To design and train an ML model using geospatial and hospital metadata features to accurately identify the nearest suitable hospital, achieving a minimum matching accuracy of 90% when compared against manual expert selection.
- To implement a Firebase-based real-time data transmission pipeline that delivers patient health parameters and GPS location to the identified hospital's emergency dashboard within 2 seconds of dispatch decision.
- To develop an automated ambulance dispatch notification module that sends a structured dispatch request to the hospital's emergency services interface, including patient vitals, current location, and estimated route.
- To evaluate the system's end-to-end performance under simulated emergency scenarios, measuring response initiation time, hospital identification accuracy, and dispatch notification reliability.
- To ensure the system operates reliably under low-connectivity conditions prevalent in semi-urban and rural Sri Lanka, incorporating offline proximity caching and fallback dispatch protocols.
- To evaluate routing accuracy of the KNN algorithm against ground-truth shortest distances across a test dataset of Sri Lankan hospital locations.

3. METHODOLOGY

The research follows a Design Science Research (DSR) methodology, which is appropriate for projects that produce an IT artifact in this case, an intelligent dispatch system with measurable real-world impact. The DSR cycle comprises problem identification, solution design, artifact development, evaluation, and knowledge contribution [17].

3.1 Research Approach

The Design Science Research (DSR) framework guides this component. The research cycle proceeds through five phases:

- Phase 1: Problem Awareness: Literature review and contextual analysis of Sri Lanka's emergency response landscape to identify specific technical and operational gaps (Completed, documented in Section 1).
- Phase 2: Suggestion: Conceptual design of the ML dispatch model, GPS integration architecture, and hospital notification pipeline.
- Phase 3: Development: Iterative implementation using Agile sprints. Each sprint delivers a testable sub-component: GPS module, ML model, Firebase integration, dispatch interface.
- Phase 4: Evaluation: Simulation-based testing using historical GPS traces and synthetic patient health event data.
- Phase 5: Conclusion: Performance benchmarking, documentation, and knowledge contribution.

3.2 Technical Methods and Algorithms

The nearest hospital identification module will evaluate two candidate ML algorithms:

Random Forest Classifier: A supervised ensemble learning algorithm that constructs multiple decision trees and aggregates predictions. Input features include Euclidean distance from patient GPS coordinates to hospital coordinates, hospital type (general/specialist/private), lastknown capacity category (low/medium/high, sourced from a static hospital registry), and road distance estimate from mapping API. Random Forest was selected based on its demonstrated superior performance over KNN in multi-criteria classification tasks [8].

K-Nearest Neighbors (KNN): Used as a baseline comparator algorithm. KNN identifies the K hospitals geographically closest to the patient's GPS location and ranks them by proximity. This serves as the fallback algorithm in low-connectivity scenarios where road distance API calls cannot be resolved.

The model will be trained on a dataset constructed from publicly available Sri Lanka hospital registry data (Ministry of Health) and synthetic GPS event data generated from documented emergency incident locations in Colombo, Gampaha, and Kandy districts.

The Haversine formula will be used for great-circle distance calculation between the patient GPS coordinate and each hospital in the registry:

$$d = 2r \times \arcsin(\sqrt{(\sin^2((\phi_2 - \phi_1)/2) + \cos(\phi_1)\cos(\phi_2)\sin^2((\lambda_2 - \lambda_1)/2))})$$

Where r is Earth's radius (6,371 km), ϕ is latitude, and λ is longitude. This formula provides accurate distance calculations for the relatively short inter-hospital distances (< 50 km) relevant to this application.

3.3 Tools and Platforms with Justification

Python (Flask/FastAPI)	Backend dispatch service. Flask is lightweight and well-suited for REST API development in resource-constrained deployments. FastAPI is used for high-performance asynchronous endpoints.
scikit-learn	ML model training and inference (Random Forest, KNN). Mature, well-documented library with extensive support for geospatial feature engineering.
Google Maps Platform API	Road distance calculation and hospital geocoding. Chosen for superior accuracy in Sri Lankan urban environments [12].
OpenStreetMap (OSM) + Nominatim	Fallback geospatial service for rural zones and offline caching of hospital coordinates.
Firestore Realtime Database / FCM	Real-time transmission of patient data to hospital dashboard. Chosen for sub-500ms delivery latency and reliability [11].
Android SDK (GPS Module)	Patient location acquisition from the smartwatch. Android's fused location provider combines GPS, Wi-Fi, and cell tower data for highest accuracy.
PostgreSQL + PostGIS	Hospital registry storage with geospatial indexing for fast nearest-neighbor queries.
Docker	Containerization of the dispatch API for scalable and reproducible deployment.
React (Web)	Hospital emergency dashboard frontend for real-time patient data visualization.

Table 2: Tools and Platforms

3.4 Data Collection and Ethical Considerations

Hospital Registry Data: The Sri Lanka Ministry of Health's publicly available hospital directory will be used to populate the hospital metadata database (name, location coordinates, type, specialization, district). No patient data from real hospitals will be accessed during development.

Synthetic Emergency Data: Patient health event data used for ML training and testing will be synthetically generated using parameters derived from published clinical guidelines for cardiac events, stroke, and respiratory distress. No real patient data will be collected or used.

GPS Simulation: Emergency GPS traces will be simulated using recorded driving routes in Colombo, Kandy, and Gampaha districts to reflect realistic ambulance dispatch scenarios.

Ethical Considerations: This project does not involve human subjects during the development phase. A formal ethics clearance application will be submitted to the SLIIT Ethics Committee prior to any pilot deployment involving real users. All location data generated during testing will be anonymized and stored locally on encrypted drives. GDPR-aligned data minimization principles will be applied to all data handling procedures.

3.5 Validation Strategy and Performance Metrics

The system will be validated through simulation-based testing and component-level unit testing. The following performance metrics will be measured:

End-to-End Response Time	Time from receipt of alert to dispatch notification sent. Target: < 10 seconds.
Hospital Identification Accuracy	Percentage of cases where the ML model's top-1 recommendation matches the expert-selected optimal hospital. Target: $\geq 90\%$.
GPS Acquisition Latency	Time to retrieve patient coordinates from Android GPS module. Target: < 2 seconds.
Firebase Notification Delivery Rate	Percentage of dispatch notifications successfully delivered to hospital dashboard. Target: $\geq 99\%$.
Model Precision / Recall / F1	Standard ML performance metrics evaluated via 5-fold cross-validation on the synthetic hospital selection dataset.
Offline Fallback Success Rate	Percentage of emergency events correctly handled using the KNN fallback when API connectivity is unavailable. Target: $\geq 85\%$.

Table 3: Validation Strategy and Performance Metrics

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

SafeBand is an end-to-end IoT and Machine Learning platform designed to continuously monitor patient health parameters, detect life-threatening emergencies, and autonomously coordinate emergency response. The system integrates wearable hardware, cloud services, mobile applications, and hospital-facing dashboards into a cohesive pipeline.

Core Components:

	Component	Role	Technology
1	ML-Driven Emergency Detection System	Analyses vitals (SpO2, BP, HR) in real-time; classifies emergency vs. normal state.	Firebase, ML classifier
2	ML-Based Smart Ambulance Dispatch	Receives alert → identifies nearest hospital → dispatches ambulance → share patient data.	Geospatial ML, AWS, Firebase
3	ML-Driven Ward Monitoring System	Continuous in-ward patient monitoring; abnormality detection for nurses/doctors.	Firebase, Hospital Dashboard
4	ML-Driven Prediction System	Predicts future deterioration; provides early warnings to clinical staff.	ML models, AWS Web App

Table 4: Components

Supporting Technologies

- SmartWatch – Collects SpO2, Heart Rate, Blood Pressure, and GPS location every few seconds.
- Firebase Realtime Database – Acts as the central data bus between the smartwatch, Android app, and ML components.
- Android Application – Patient-facing mobile app; displays health data and emergency alerts.
- AWS (Amazon Web Services) – Hosts the web application for doctor/admin access.
- Hospital Dashboards – Emergency dashboard (ambulance dispatch view) and ward dashboard

4.2 System Diagram of Components

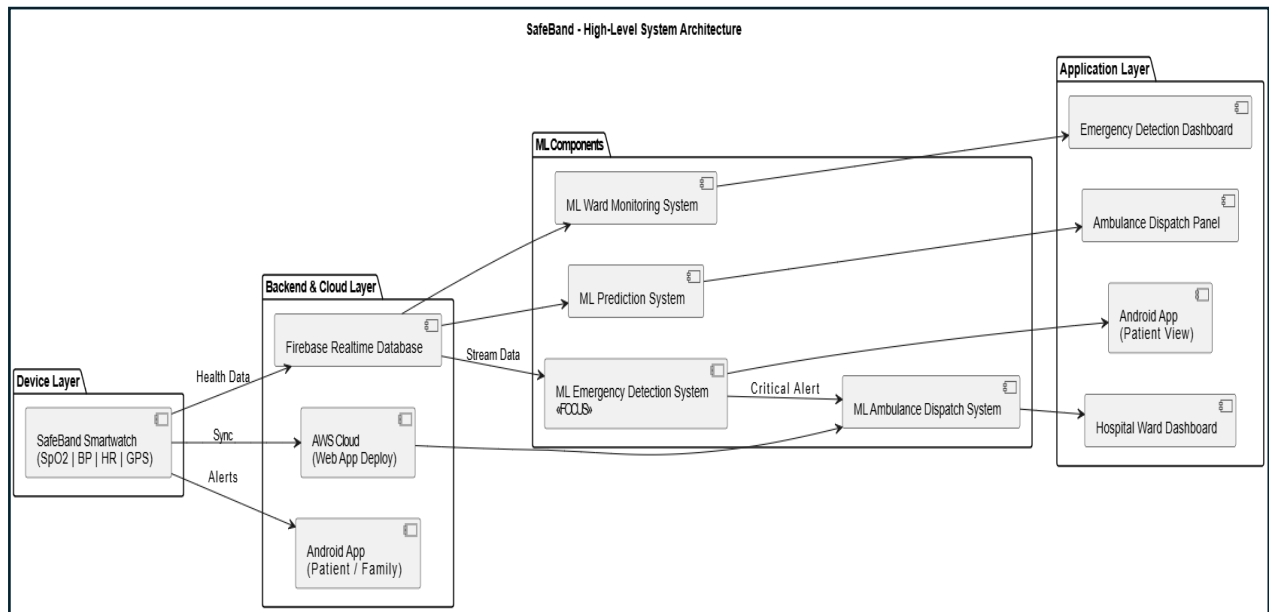


Figure 1: System Diagram

4.3 Highlight the individual component and its parts

This component is the focal point of the research contribution. It sits between the Emergency Detection System and the Hospital Infrastructure, acting as the intelligent routing and dispatch engine.

Sub-Components

Emergency Alert Receiver	Listens to Firebase for EMERGENCY_ALERT events from the Emergency Detection System. Parses the alert payload including patient ID, vitals snapshot, GPS coordinates, and timestamp.
Hospital Proximity Engine	Uses patient GPS coordinates alongside the hospital database to compute real-time distances. Applies an ML scoring model to rank hospitals by proximity, available capacity, and specialization match.
Ambulance Dispatch Controller	Issues a dispatch command to the nearest available ambulance via the hospital dashboard API. Confirms dispatch and calculates estimated time of arrival (ETA) using live traffic data.
Patient Data Relay	Pushes the complete patient record (vitals history, emergency type, and GPS coordinates) to the receiving hospital's Emergency Dashboard, ensuring clinical staff are fully briefed before the patient arrives.
Status Tracker	Monitors the ambulance GPS in real-time and continuously updates all stakeholder dashboards (hospital emergency dashboard and Android app) with live ETA and current ambulance location until patient handover.

Table 5: individual component

4.4 Component interactions

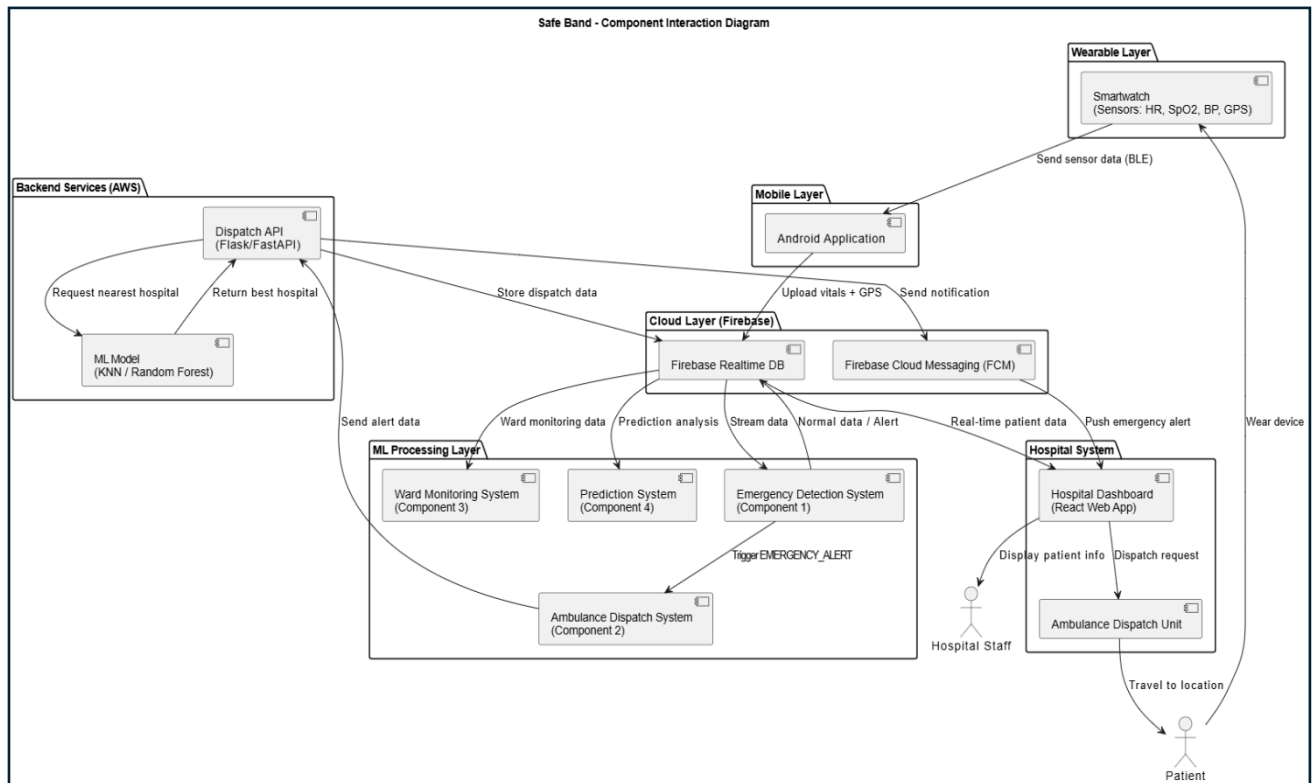


Figure 2: Component interactions

This component interaction diagram illustrates how the Safe Band system components communicate during an emergency. The smartwatch continuously collects patient health data and sends it to the Firebase cloud via the Android application. The Emergency Detection System analyzes this data and triggers an alert when a critical condition is detected. The Ambulance Dispatch System then processes the alert, uses a machine learning model to identify the nearest hospital, and sends the information through the backend API. The hospital dashboard receives real-time patient data and notifications, enabling staff to dispatch an ambulance immediately and monitor the situation effectively.

4.5 Data flow explanation

The Flow diagram illustrates how the system handles an emergency scenario from start to finish. The smartwatch continuously collects raw sensor data including SpO2, blood pressure, heart rate, and GPS location, which is sent through the Android app and stored in the Firebase Realtime Database. This data is then analysed by the ML-Based Emergency Detection System to determine whether an emergency exists. If no emergency is detected, the system continues monitoring the patient normally. However, if an emergency is detected, the system extracts the patient's GPS coordinates and vital signs, triggering the ML-Based Ambulance Dispatch component, which calculates the nearest hospital, prepares a complete data payload containing the patient ID, vitals, location, and emergency type, and sends it to the AWS Hospital Dashboard. An ambulance is then immediately dispatched to the patient's location, while the system loops back to Firebase to continue recording data throughout the entire emergency response process.

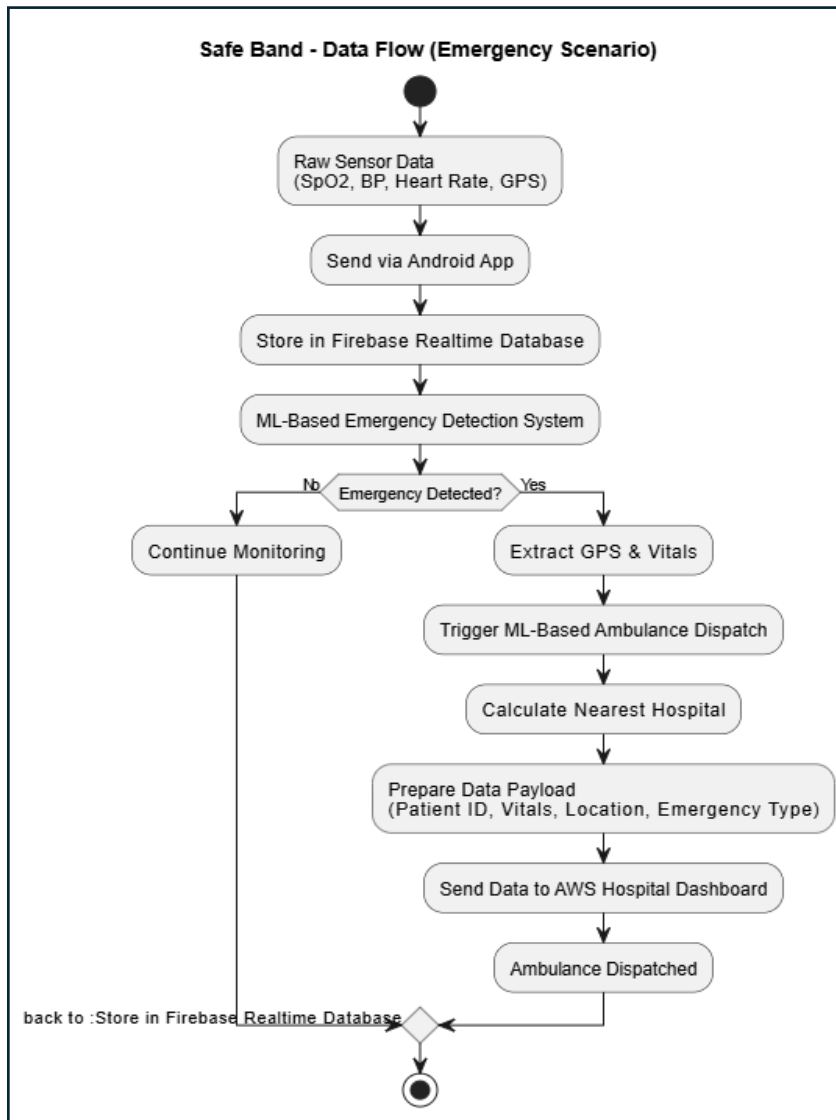


Figure 3: Data flow

4.6 Scalability and security considerations

Scalability:

The system is designed to efficiently handle increasing workloads and emergency situations through scalable technologies. Firebase Realtime Database uses horizontal sharding by region to ensure low-latency data access across different geographic locations. AWS Lambda with auto-scaling capabilities allows the system to handle sudden spikes in emergency events without manual provisioning. The Android application uses background synchronization with Work Manager to enable continuous monitoring while maintaining battery efficiency. Hospital databases are supported with read replicas on AWS RDS to ensure fast data retrieval even under high concurrent usage. Additionally, ambulance dispatch requests are managed using queue-based processing (Amazon SQS) to prevent data loss during peak loads, while AWS CloudFront CDN ensures fast and reliable dashboard access for hospital staff across Sri Lanka.

Security Considerations:

The system incorporates strong security measures to protect sensitive healthcare data and ensure compliance with industry standards. All data transmissions are secured using TLS 1.3 encryption, while Firebase Authentication with JWT tokens and role-based access control ensures secure user access. Patient data privacy is maintained through anonymization and encryption of personally identifiable information, aligning with GDPR and HIPAA standards. Access to Firebase is restricted to authenticated applications, and AWS IAM roles are used to manage permissions securely without exposing root credentials. The ambulance dispatch API protects HMAC-signed requests and regular key rotation, following a zero-trust approach. Furthermore, all system activities are logged using AWS CloudTrail and Firebase Audit Logs to ensure accountability and traceability.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Patient / Wearable User	Primary end-user wearing the Safe Band smartwatch. Requires automatic emergency activation without manual intervention.
Hospital Emergency Staff	Receives real-time patient data on the hospital dashboard. Requires fast, accurate, and actionable information.
Ambulance Dispatch Operator	Receives automated dispatch notifications. Requires clear location data and incident details.
Hospital Administrator	Oversees incident logs, response time analytics, and system configuration.

Table 6: Stakeholder Identification

5.2 Functional Requirements

- The system shall receive a CRITICAL event alert from the Emergency Detection System via a REST API endpoint within 1 second of event classification.
- The system shall acquire the patient's real-time GPS coordinates from the Android smartwatch within 2 seconds of receiving the critical alert.
- The system shall identify the nearest suitable hospital from the registry using the ML algorithm and return the top recommendation within 3 seconds.
- The system shall transmit a complete patient health data payload (biometrics, GPS, event type) to the identified hospital's Firebase node within 2 seconds of hospital selection.
- The hospital emergency dashboard shall display the incoming patient data in real-time without requiring manual refreshment.
- The system shall send an ambulance dispatch notification to the hospital emergency team via Firebase Cloud Messaging and SMS fallback.
- The system shall log all incident data (patient ID, GPS, hospital assigned, dispatch time) to the central database for audit and analytics.
- The system shall activate the KNN fallback algorithm if the Random Forest model confidence score is below 0.75 or if the API call to the mapping service fails.
- The system shall support a hospital registry of at least 500 hospitals across Sri Lanka, with geospatial indexing for sub-100ms query response.
- The system will generate a unique incident ID for each dispatch event and provide a tracking interface for the hospital administrator.

5.3 Non-Functional Requirements

Performance	End-to-end response time (alert received → dispatch triggered) must be under 10 seconds under normal network conditions.
Reliability	System uptime must be $\geq 99.5\%$. Firebase ensures high availability. Fallback dispatch via SMS ensures operation during API outages.
Security	All patient data encrypted in transit (TLS 1.3) and at rest (AES-256). JWT authentication for all API endpoints. RBAC for hospital dashboard

	access.
Scalability	The dispatch API must handle up to 1,000 concurrent emergency events without performance degradation, using horizontal Docker container scaling.
Usability	The hospital emergency dashboard must be usable by medical staff with minimal training. Key information (patient location, vitals, severity) must be visible within 3 seconds of page load.
Portability	The backend dispatch service must be deployable on any Dockercompatible cloud provider (AWS, GCP, or Azure) without code modification.
Maintainability	ML model retraining must be executable via a single automated script. Hospital registry updates must be possible without system downtime.
Regulatory Compliance	All data handling must comply with Sri Lanka's Personal Data Protection Act (PDPA) 2022 and the system must be compatible with the Ministry of Health's eHealth framework.

Table 7: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary Market: Private and semi-government hospitals in Sri Lanka (approximately 110 private hospitals registered with the Private Hospitals Act). Secondary Market: Insurance companies offering health insurance products who can bundle Safe Band as a value-added service. Tertiary Market: Corporate health & safety programs for large organizations with aging workforces.

6.2 Value Proposition

- For hospitals: Real-time pre-arrival patient health data enables faster and more effective emergency preparation, reducing in-hospital response time and potentially improving survival outcomes.
- For patients/insurers: Fully automated, wearable-triggered emergency dispatch removes the need for manual SOS activation critically when the patient is unconscious or unable to call for help.
- For ambulance services: Automated dispatch requests with verified GPS location and health data reduce operator workload and improve dispatch accuracy.

6.3 Revenue Model

- **Hospital SaaS Subscription:** Monthly/annual subscription for access to the hospital dashboard and dispatch notification API. Tiered by hospital size (Small: LKR 15,000/month, Medium: LKR 35,000/month, Large: LKR 75,000/month).
- **Per-Device Licensing:** Annual license fee of LKR 2,500 per Safe Band device registered to the hospital's dispatch network.
- **Insurance Partnership Model:** Revenue-share agreement with insurance companies that bundle Safe Band with health insurance policies.
- **API Access for Third-Party Integration:** Paid API access for third-party hospital management systems to integrate with the dispatch notification service.

6.4 Cost Estimation

Initial development costs are estimated at LKR 850,000 (detailed in Chapter 7). Annual operational cost is estimated at LKR 240,000 (cloud hosting, API subscriptions, maintenance). Break-even is projected at 8 hospital subscriptions, achievable within 6 months of commercial launch based on the identified target market of 110 private hospitals.

6.5 Competitive Advantage

- Sri Lanka-specific design: The only known system tailored to the specific infrastructure constraints of Sri Lanka's emergency services and hospital ecosystem.
- No dependency on real-time hospital APIs: Unlike global competitors, the system operates effectively without live bed-availability data, making it deployable immediately.
- Fully automated pipeline: Zero manual intervention required from patients to dispatch a key differentiator over existing wearable emergency SOS systems.

7. BUDGET AND JUSTIFICATION

Item	Category	Estimated Cost (LKR)	Justification
Android Smartwatch	Hardware	60,000	Primary data collection device for GPS and sensor integration testing
AWS EC2 t3.medium (12 months)	Cloud Services	120,000	Hosting the dispatch API backend and Firebase project
Google Maps Platform API (12 months, dev tier)	Cloud Services	18,000	Road distance calculation and hospital geocoding
Firebase Blaze Plan (12 months)	Cloud Services	15,000	Real-time database and Cloud Messaging for hospital notifications
Python / scikit-learn / Flask (open source)	Software	0	No license cost — all tools are open source
Docker Desktop Pro (1 developer, 12 months)	Software	12,000	Containerization and local deployment testing
TOTAL		225,000	

Table 8: Budget and Justification

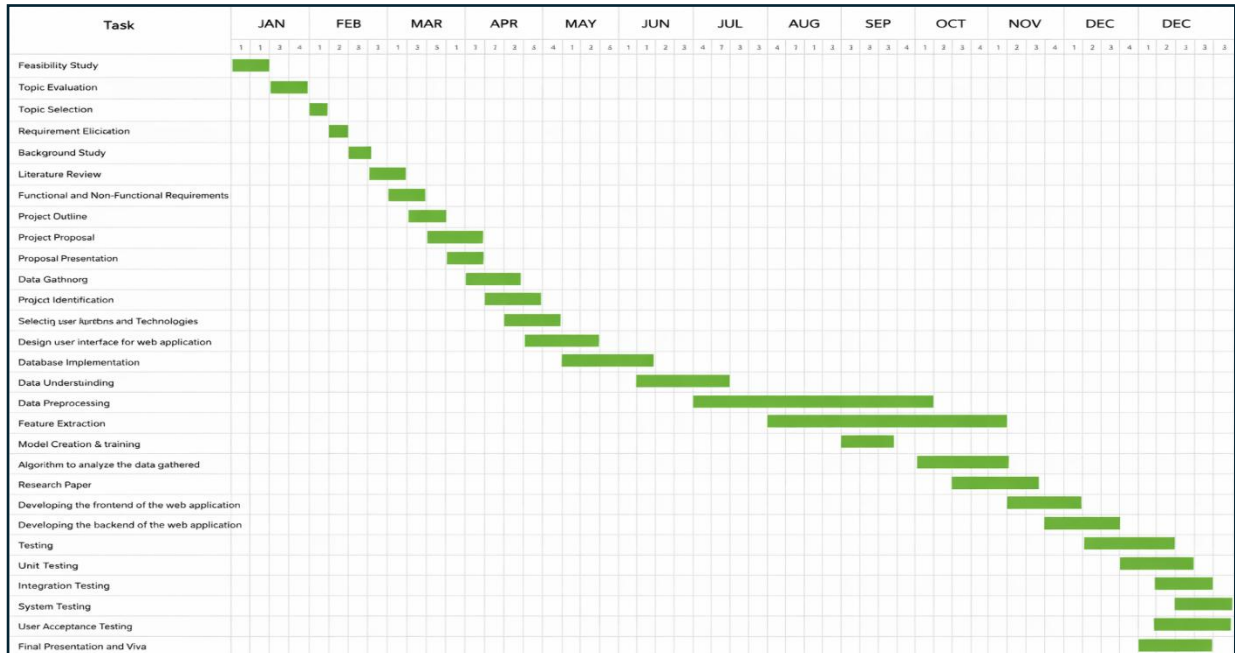
8. WORK BREAKDOWN STRUCTURE (WBS)

ID	Task	Duration	Deliverable
1.0	SEMESTER 1		
1.1	Literature Review & Problem Analysis	2 weeks	Annotated bibliography (Section 1.2)
1.2	Hospital Registry Data Collection	1 week	PostgreSQL hospital registry with 200+ Sri Lanka hospitals
1.3	Synthetic Dataset Generation	2 weeks	Labeled GPS-event dataset for ML training
1.4	ML Model Development (Random Forest)	3 weeks	Trained RF model with $\geq 90\%$ accuracy
1.5	KNN Fallback Algorithm Implementation	1 week	KNN module with offline caching
1.6	GPS Acquisition Module (Android)	2 weeks	Android GPS module integrated with smartwatch
1.7	REST API — Alert Intake Endpoint	1 week	Flask API endpoint for C3 alert receipt
1.8	Unit Testing — ML Model & GPS Module	1 week	Unit test report ($\geq 80\%$ coverage)
1.9	Semester 1 Progress Report	1 week	Interim project report
2.0	SEMESTER 2		
2.1	Firebase Integration — Realtime DB	2 weeks	Real-time patient data transmission to hospital node
2.2	Hospital Emergency Dashboard (React)	3 weeks	Functional web dashboard for hospital staff
2.3	Ambulance Dispatch Notification Module	2 weeks	FCM + SMS dispatch notification system
2.4	Incident Logging & Admin Interface	1 week	Incident database + administrator tracking view
2.5	End-to-End Integration with C3 (Emergency Detection)	2 weeks	Integrated pipeline from C3 trigger to dispatch
2.6	Simulation-Based System Testing	2 weeks	Test report: response time, accuracy, delivery rate

2.8	Final Documentation & Proposal Report	1 week	Completed research proposal and technical report
2.9	Final Presentation	1 week	Viva presentation and demonstration
2.7	Performance Optimization	1 week	Optimized system

Table 9: Work Breakdown Structure

9. GANTT CHART



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11.APPENDICES

- **Preliminary UI mockups**

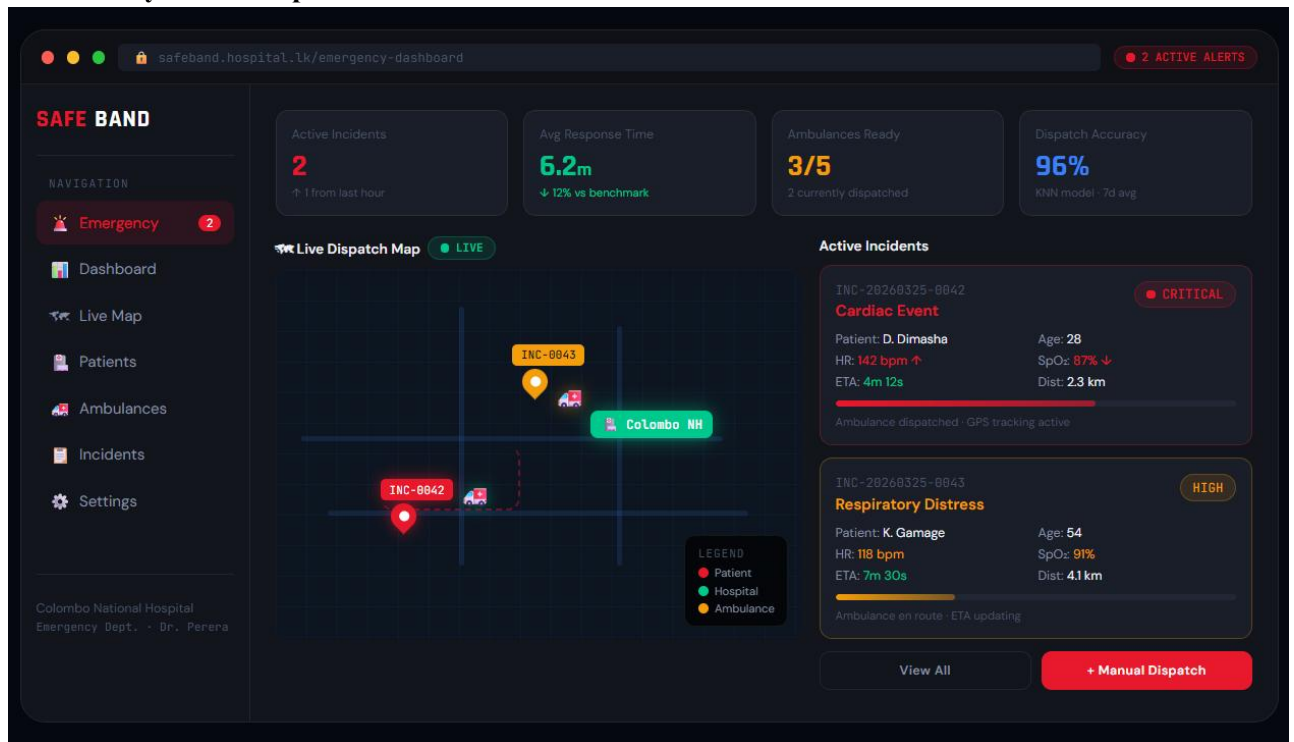


Figure 4: Preliminary UI

- **Risk analysis**

The risk register for the ML-Based Smart Ambulance Dispatch System outlines a comprehensive assessment of key technical, operational, clinical, and regulatory risks associated with the system's deployment. Conducted in alignment with the ISO 31000:2018 Risk Management framework, the analysis identifies potential threats, evaluates their likelihood and impact, and defines appropriate mitigation strategies. Several high-severity technical risks are highlighted, including GPS inaccuracies in low-signal environments, Firebase Realtime Database outages affecting real-time communication, and incorrect hospital identification by the KNN model due to outdated or anomalous data. Infrastructure-related challenges, particularly poor mobile connectivity in rural areas of Sri Lanka, also pose a significant threat to system reliability. From a clinical and operational perspective, false-positive emergency alerts and limited staff familiarity with the dashboard interface may lead to inefficiencies and reduced trust in the system. Additionally, data-related risks such as incomplete hospital registry information can negatively impact decision-making accuracy. Security concerns are critical, with unauthorized access potentially violating Sri Lanka's PDPA 2022 regulations. Other notable risks include smartwatch battery depletion during emergencies and possible delays in obtaining regulatory approvals for pilot deployment. Overall, the register emphasizes proactive mitigation and contingency planning to ensure system robustness, reliability, and compliance in real-world healthcare environments.

